THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

# ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** energy | Main source. |
| 2 | **(b)** much sunlight | High albedo. |
| 3 | **(c)** carbon dioxide | $CO_2$. |
| 4 | **(d)** atmosphere | Gases. |
| 5 | **(b)** absorb sunlight | Low albedo. |
| 6 | **(d)** carbon dioxide | $CO_2$. |
| 7 | **(a)** geosphere | Solid. |
| 8 | **(b)** hotter than the poles | Direct rays. |
| 9 | **(d)** more $CO_2$ in air | Fewer trees. |
| 10 | **(c)** hydrosphere | Water. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Geosphere and hydrosphere (1); Atmosphere and biosphere (1). | 1+1 |
| 12 | Change of water into vapour (2). | 2 |
| 13 | Fraction of sunlight reflected by a surface (2). | 2 |
| 14 | Trapping of heat by gases in the atmosphere (2). | 2 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Fossil fuels (1); Deforestation (1); Agriculture and waste (1). | 1+1+1 |
| 16 | Save water (1); Save energy (1); Avoid plastics (1). | 1+1+1 |
| 17 | Absorb $CO_2$ (1); Give oxygen (1); Rain (1). | 1+1+1 |
|  | **OR:** Melting ice (1); Rising sea level (1); Extreme weather (1). | 1+1+1 |
| 18 | Solar cooker (1); Water heater (1); Solar cells (1). | 1+1+1 |
|  | **OR:** Warming (1); Climate change (1); Sea-level rise (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Geosphere (1); Hydrosphere (1); Atmosphere (1); Biosphere (1); Interactions (1). | 1+1+1+1+1 |
|  | **OR:** Source (1); Speed (1); Distribution (1); Reflection (1); Use (1). | 1+1+1+1+1 |
| 20 | Gases (1); Trap heat (1); Natural (1); Enhanced (1); Warming (1). | 1+1+1+1+1 |
|  | **OR:** Cause (1); Gases (1); Effects (1); Control (1); Role of individuals (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** evaporation | Heat. |
| 2 | **(b)** sunlight | Solar energy. |
| 3 | **(d)** biosphere | Life. |
| 4 | **(b)** condensation | Cooling. |
| 5 | **(b)** renewable | Inexhaustible. |
| 6 | **(a)** greenhouse gas | Traps heat. |
| 7 | **(d)** precipitation | Falling water. |
| 8 | **(b)** Lifestyle for Environment | Eco-friendly. |
| 9 | **(b)** ultraviolet rays | UV. |
| 10 | **(b)** transpiration | Stomata. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Change of vapour into water droplets (2). | 2 |
| 12 | Carbon dioxide (1); Methane (1). | 1+1 |
| 13 | Rise in the Earth's average temperature (2). | 2 |
| 14 | Water falling as rain, snow or hail (2). | 2 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Skin cancer (1); Cataract (1); Damage to crops (1). | 1+1+1 |
| 16 | Snow: high (1); Forest: low (1); Ocean: low (1). | 1+1+1 |
| 17 | t = d/v (1); 500 s (1); About 8 minutes (1). | 1+1+1 |
| | **OR:** Evaporation (1); Condensation (1); Precipitation (1). | 1+1+1 |
| 18 | Provides air (1); Protects from UV (1); Keeps Earth warm (1). | 1+1+1 |
| | **OR:** Plants (1); Animals (1); Microbes (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Where (1); Function (1); CFCs (1); Effects (1); Protection (1). | 1+1+1+1+1 |
| | **OR:** Definition (1); Snow (1); Forest (1); Ocean (1); Effects (1). | 1+1+1+1+1 |
| 20 | Evaporation (1); Transpiration (1); Condensation (1); Precipitation (1); Runoff (1). | 1+1+1+1+1 |
| | **OR:** Meaning (1); Aim (1); Save energy (1); Save water (1); Reduce waste (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** energy | Electricity. |
| 2 | **(b)** stratosphere | Above. |
| 3 | **(a)** photosynthesis | Food making. |
| 4 | **(b)** water | Water. |
| 5 | **(d)** ozone layer | Chemicals. |
| 6 | **(c)** 78 per cent | Largest. |
| 7 | **(d)** avoided | Pollution. |
| 8 | **(c)** temperature | Average. |
| 9 | **(b)** pulses | Legumes. |
| 10 | **(b)** 3 x 10^8 m/s | c. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Conversion of atmospheric nitrogen into usable compounds (2). | 2 |
| 12 | Layer of ozone gas that absorbs UV rays (2). | 2 |
| 13 | Loss of water vapour by plants through stomata (2). | 2 |
| 14 | Lifestyle for Environment movement (2). | 2 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Respiration (1); Burning (1); Decay (1). | 1+1+1 |
| 16 | Equator: hot (1); Poles: cold (1); Sunlight angle (1). | 1+1+1 |
| 17 | Oceans (1); Rivers (1); Ice and groundwater (1). | 1+1+1 |
| | **OR:** Plant trees (1); Save energy (1); Use solar energy (1). | 1+1+1 |
| 18 | Fixation (1); Decay (1); Nitrification (1). | 1+1+1 |
| | **OR:** Rocks (1); Soil (1); Mountains (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Photosynthesis (1); Respiration (1); Decay (1); Burning (1); Ocean (1). | 1+1+1+1+1 |
| | **OR:** Renewable (1); Clean (1); Abundant in India (1); Uses (1); Limitations (1). | 1+1+1+1+1 |
| 20 | Fixation (1); Absorption (1); Food chain (1); Decay (1); Return (1). | 1+1+1+1+1 |
| | **OR:** Absorb CO2 (1); Give oxygen (1); Water cycle (1); Habitat (1); Soil (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.